

Series Circuit

LEDs in a Chain

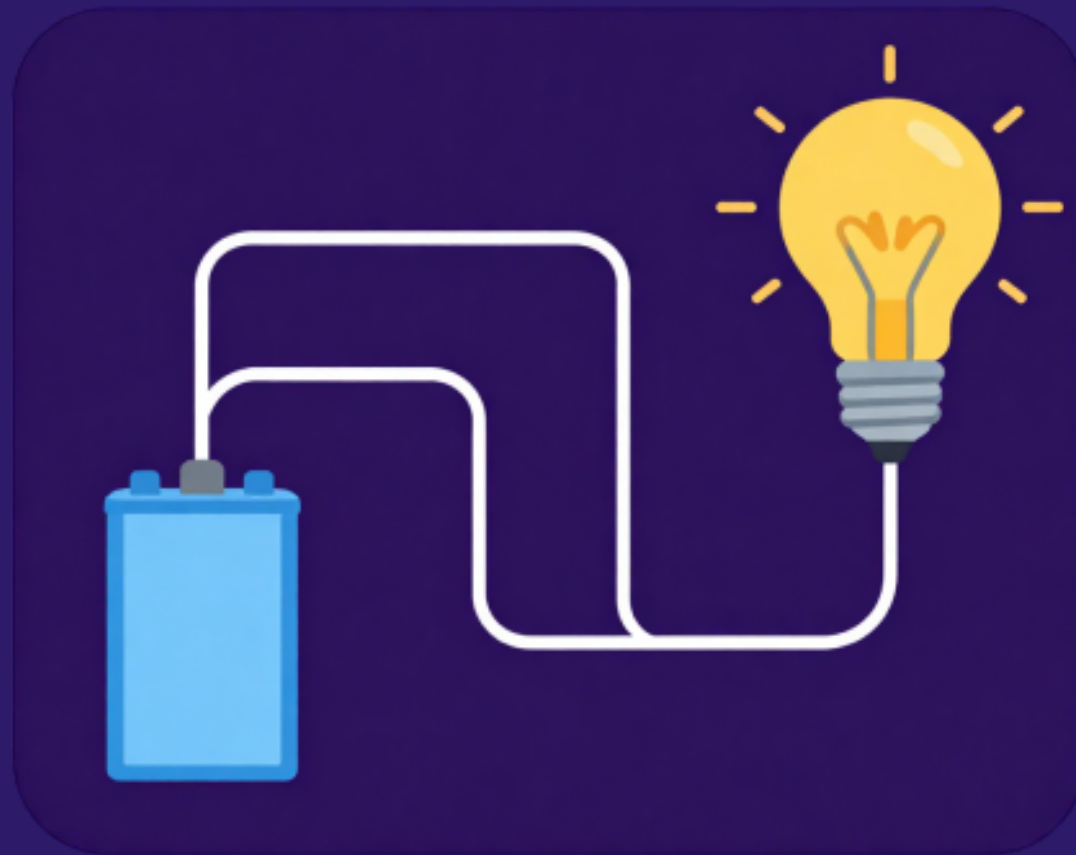
Grade 3 | Hardware



What is a Circuit?

A circuit is a complete loop that lets electricity flow from a battery, through wires and components, and back again.

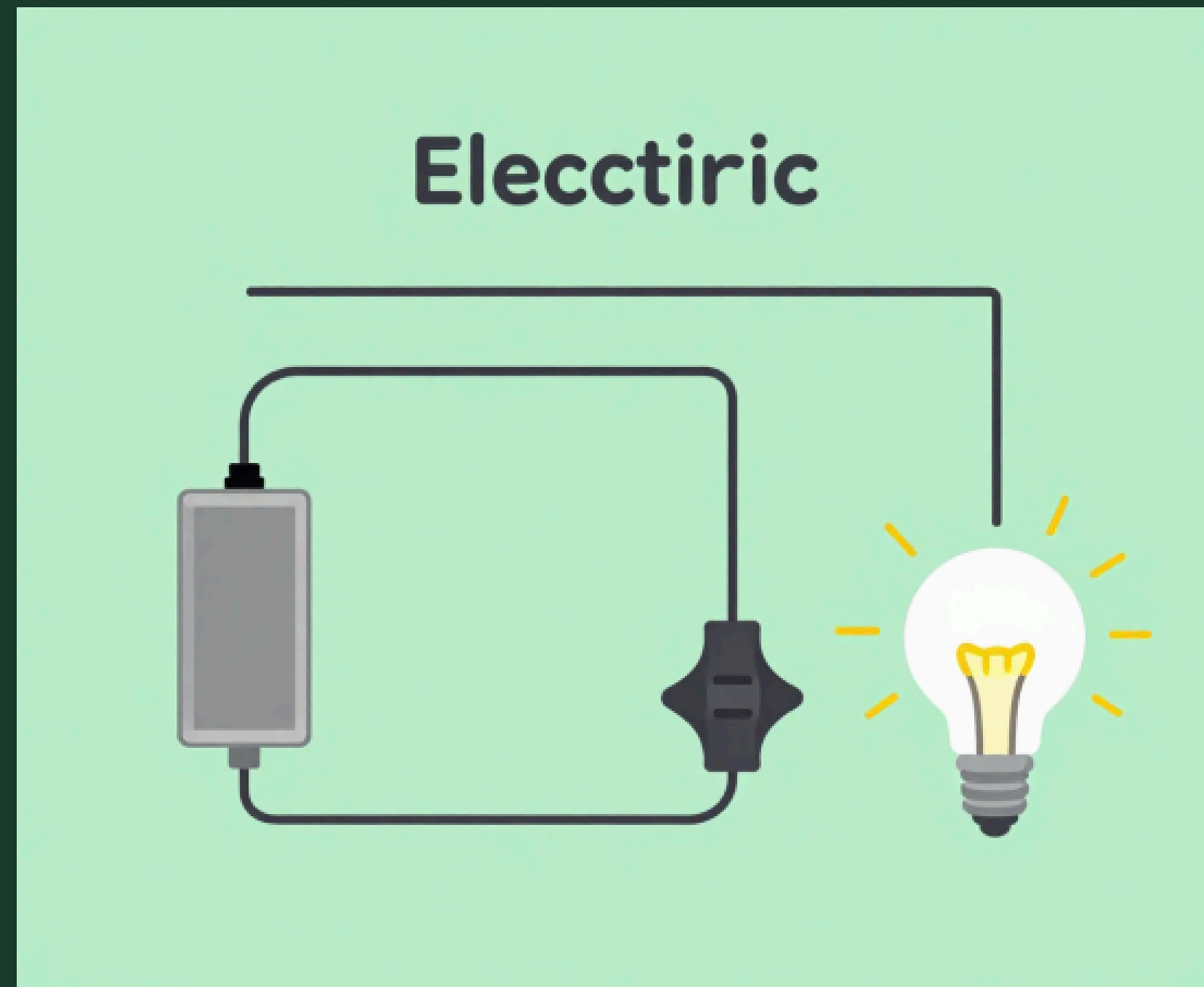
Think of it like a circular race track — the cars (electricity) must go all the way around to finish the race!



Parts of a Circuit

Every circuit needs these parts:

1. Battery — gives power (like a pump)
2. Wires — carry electricity (like pipes)
3. Switch — turns circuit ON or OFF
4. Load — the thing that uses power (like an LED or buzzer)



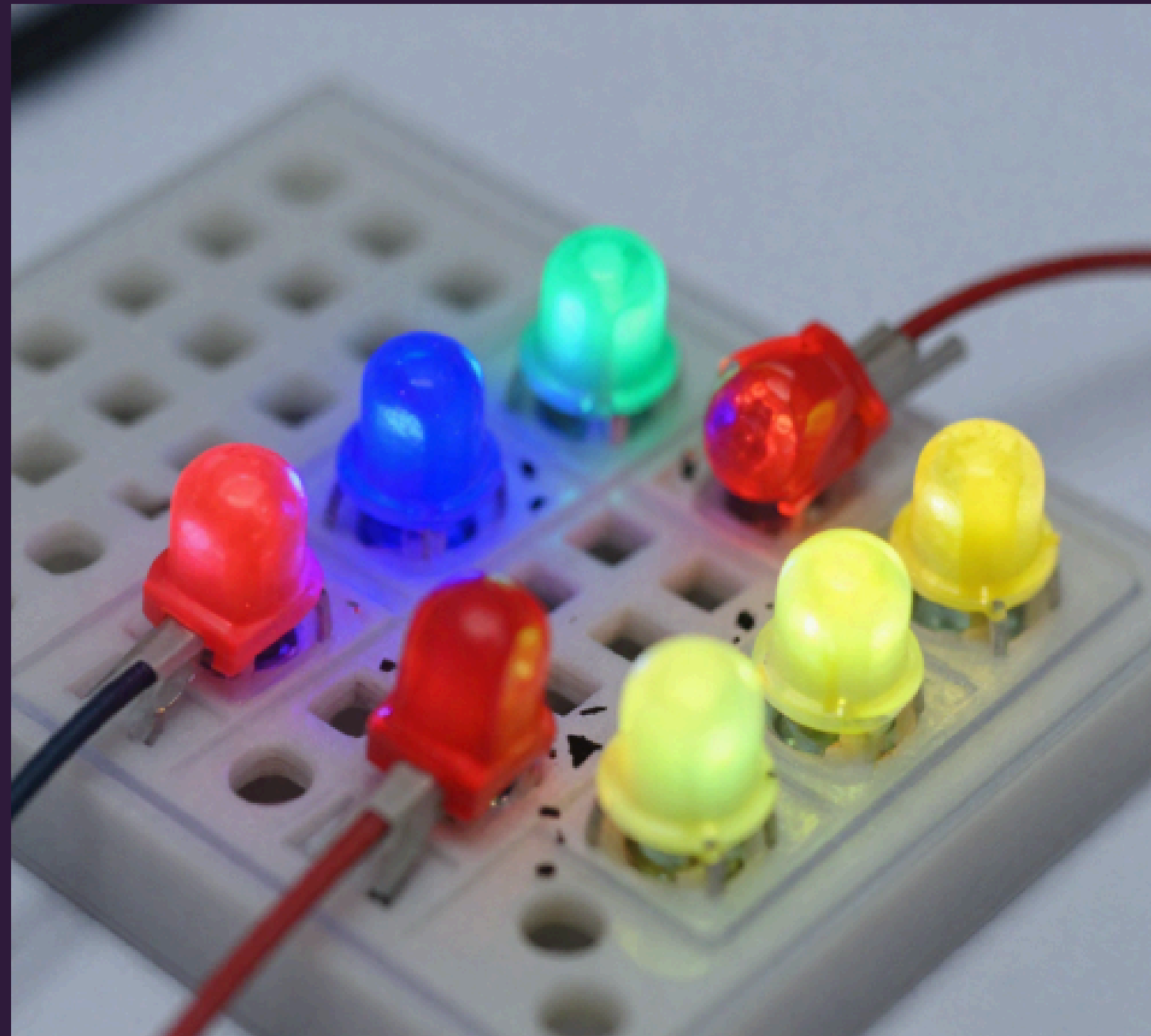
What is an LED?

LED = Light Emitting Diode

It is a tiny light that glows when electricity passes through it.

LEDs are:

- Very small and bright
- Use very little power
- Come in many colours
- Have 2 legs: long (+) and short (-)

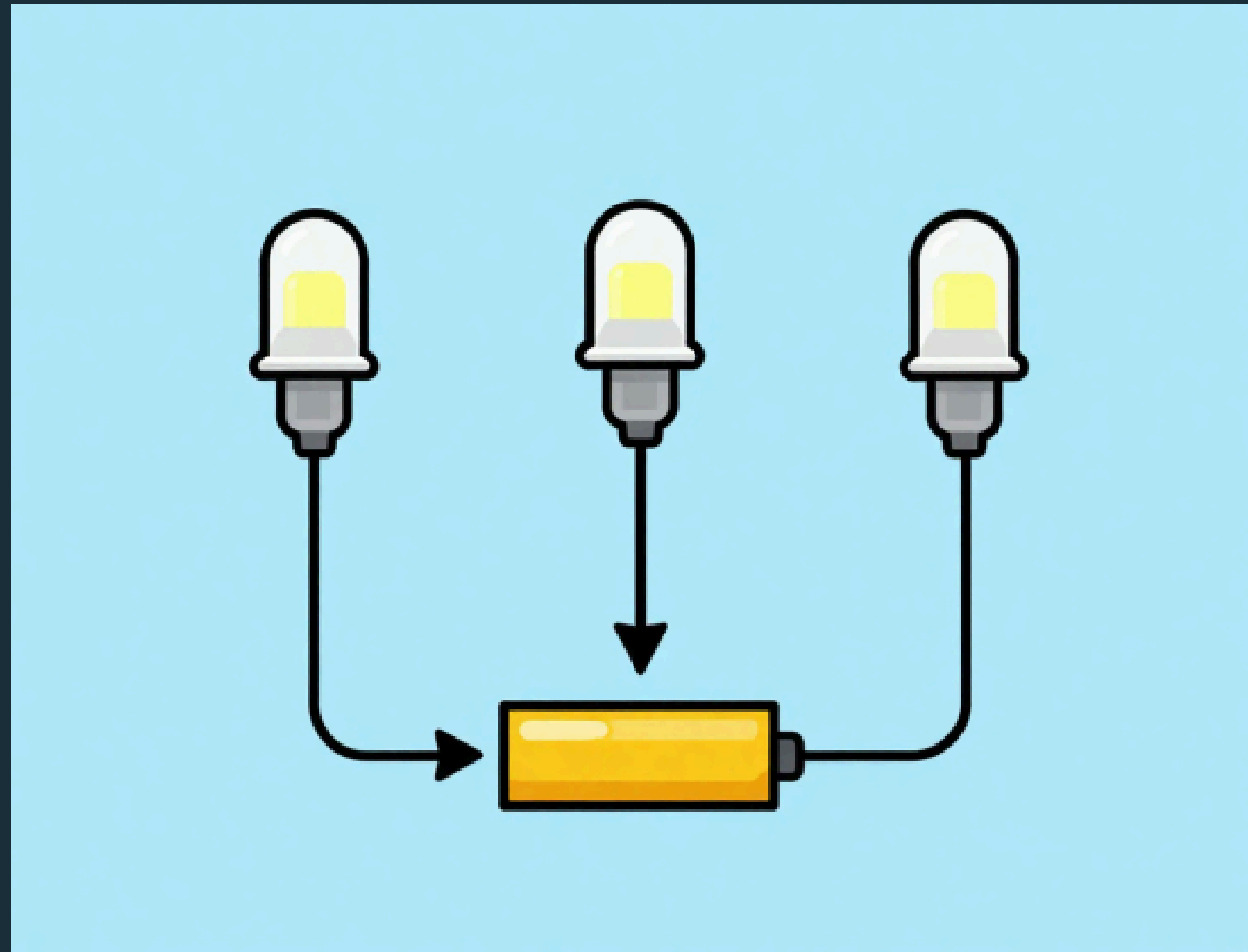


What is a Series Circuit?

In a series circuit, all parts are connected one after another in a single loop.

Electricity has only ONE path to follow.

Imagine people walking in a single-file line through a narrow hallway — everyone must go the same way!

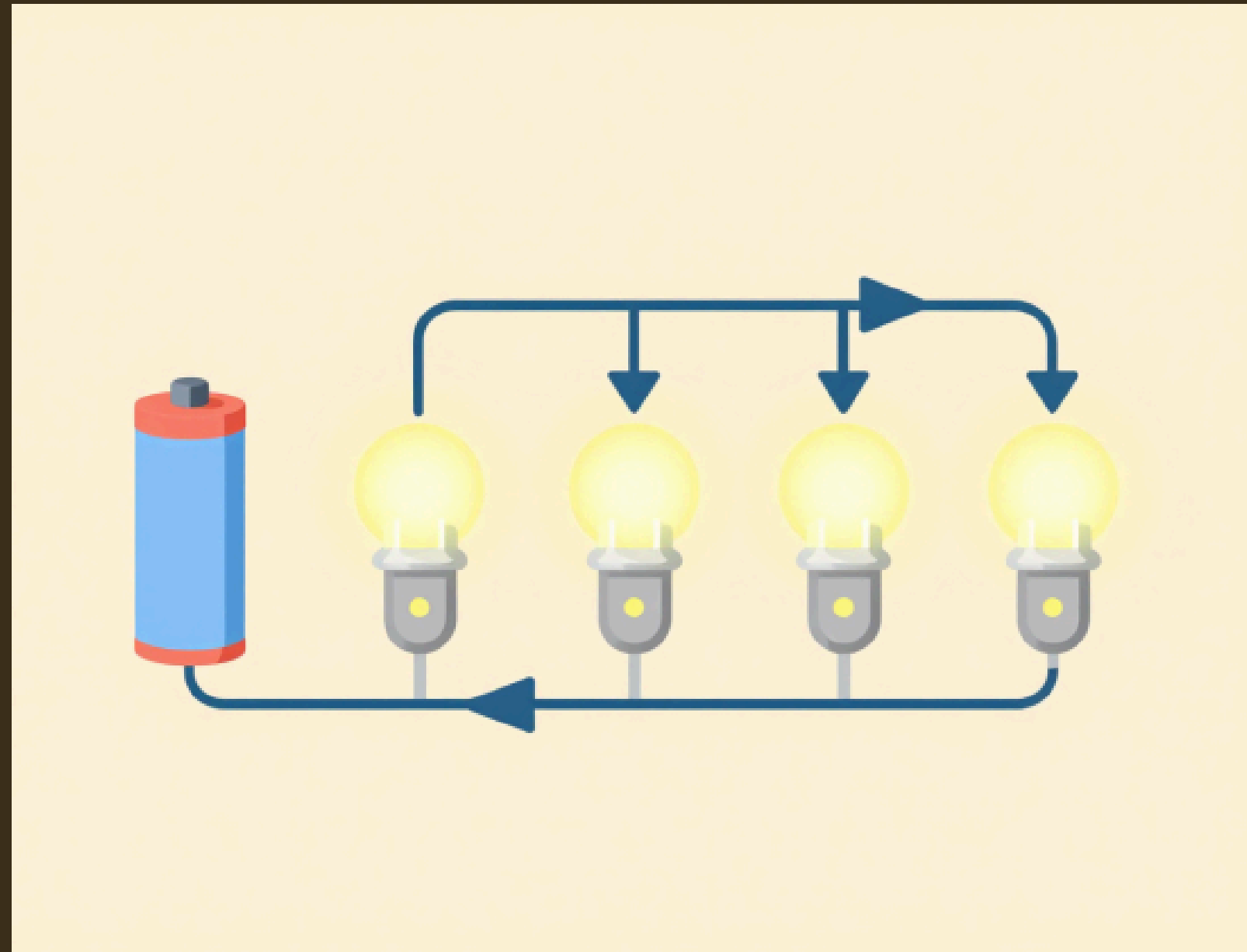


How Current Flows

Current flows from the + side of the battery, through each LED one by one, and back to the - side.

The same amount of current passes through every LED in the chain.

Think of it like water flowing through a single pipe — it visits every tap!

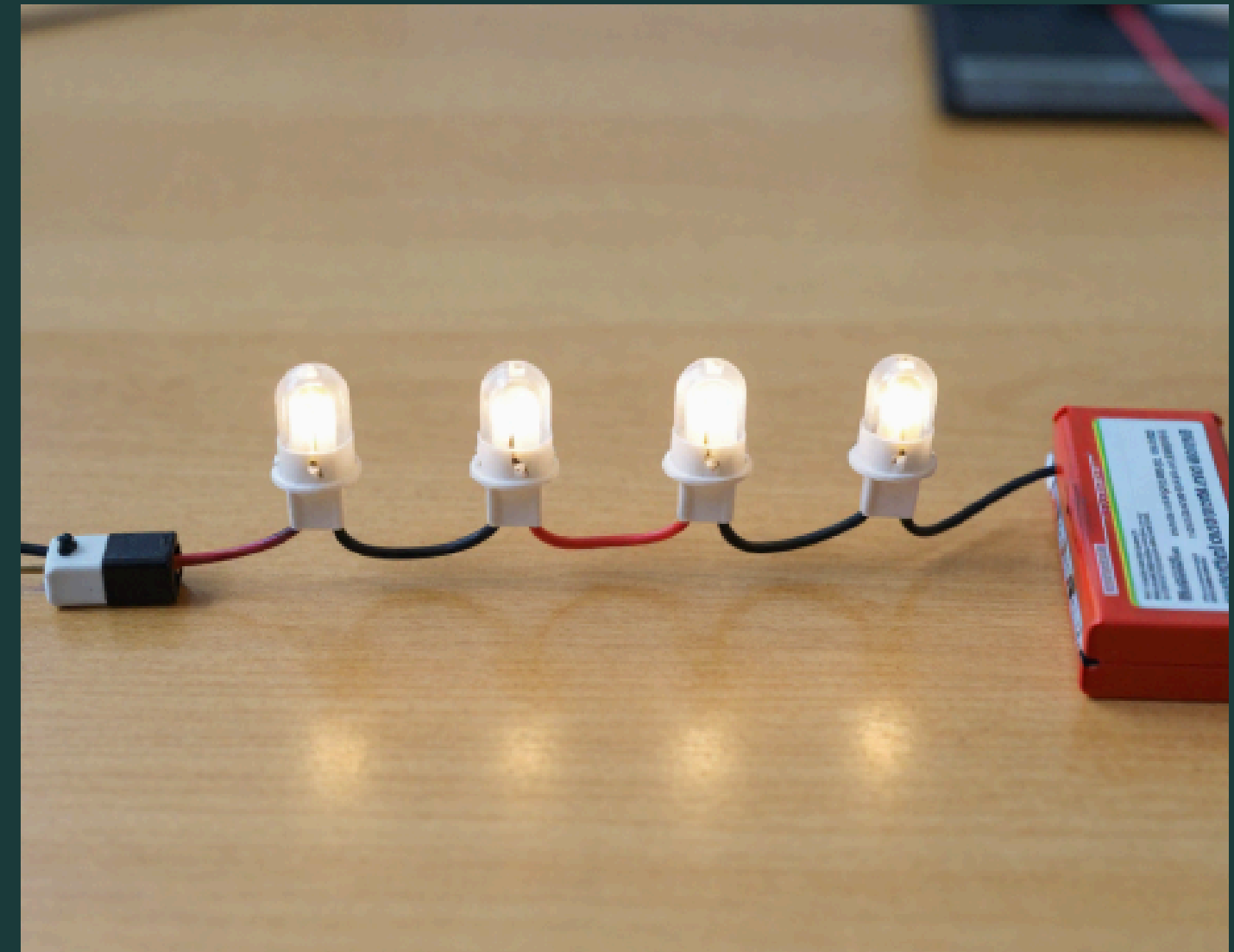


LEDs in a Chain

When we connect LEDs in series:

- They form a chain — one after another
- Each LED shares the battery voltage
- More LEDs = dimmer each one glows
- All LEDs get the same current

It is like sharing one pizza among more friends — each gets a smaller slice!



One Goes Out, All Go Out!

In a series circuit, there is only ONE path.

If one LED breaks or is removed:

- The chain is broken
- No electricity can flow
- ALL the LEDs turn off!

Think of it like a train track —
if one piece is missing, the train stops!

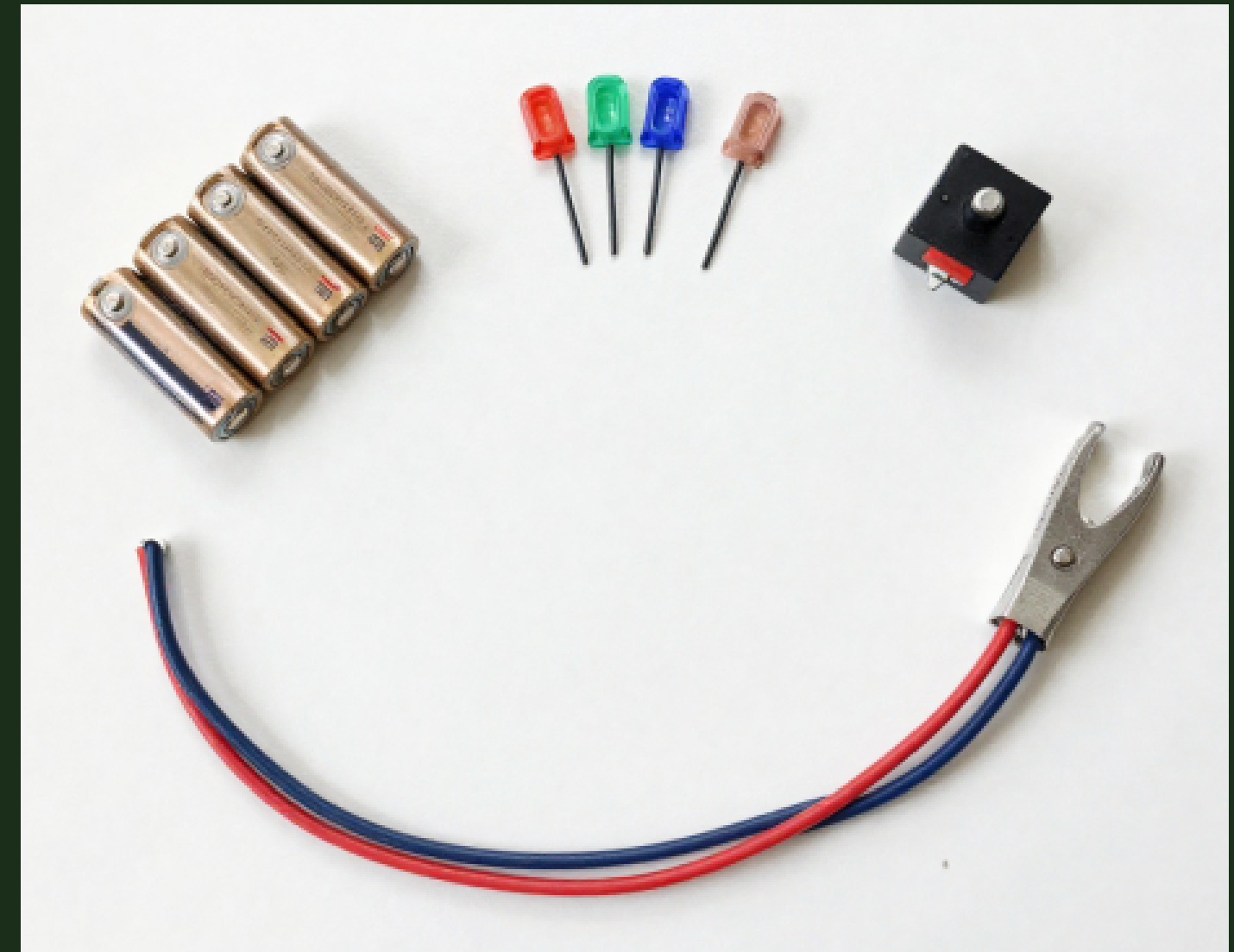


Let's Build It!

Materials you need:

- 1 battery (3V coin cell or AA)
- 3 LEDs (any colour)
- Connecting wires or crocodile clips
- 1 small switch (optional)
- Tape to hold wires

Ask your teacher before starting!



Step-by-Step Build

Follow these steps:

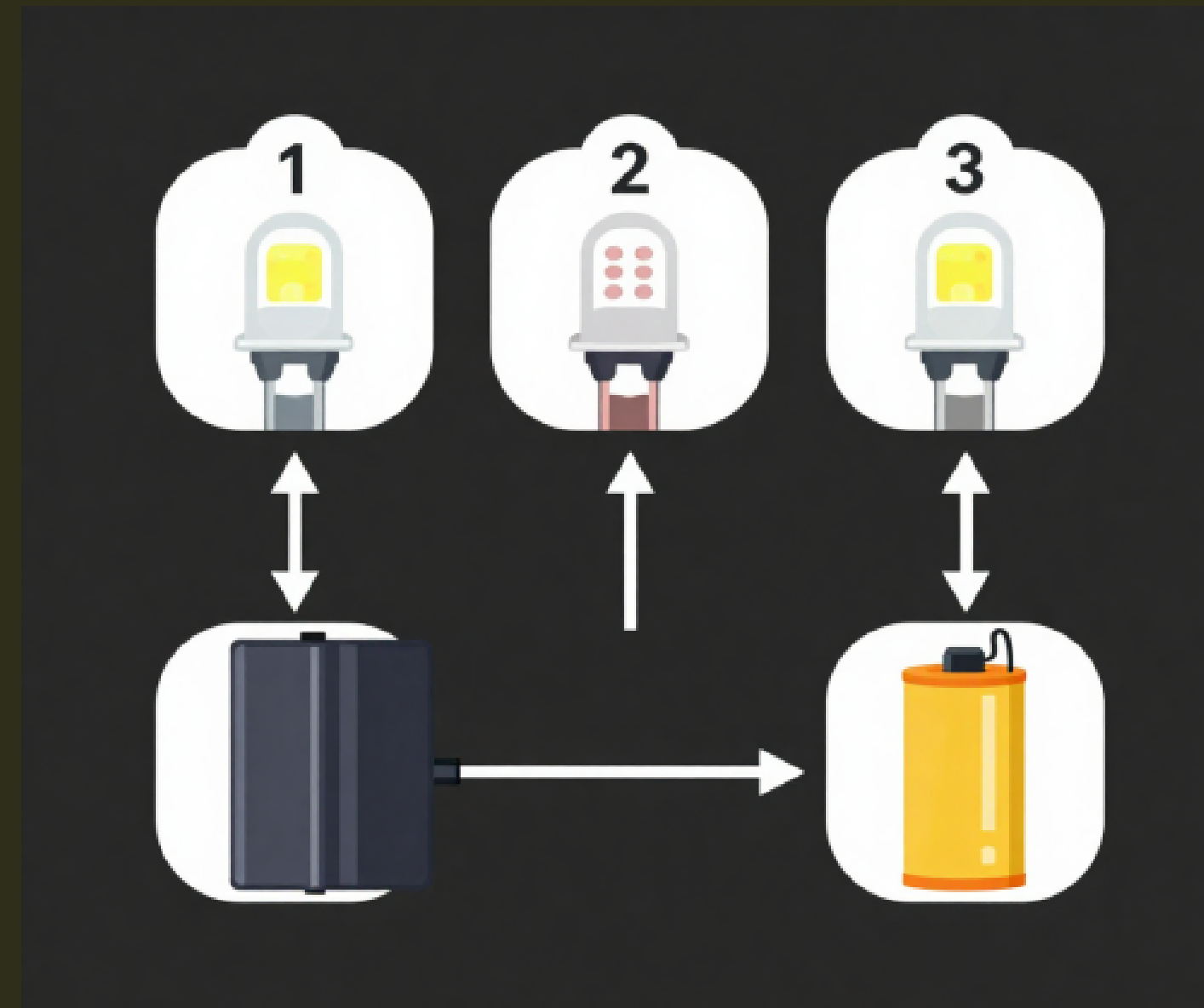
Step 1: Connect wire from battery (+) to
LED 1 long leg (+)

Step 2: Connect LED 1 short leg (-) to
LED 2 long leg (+)

Step 3: Connect LED 2 short leg (-) to
LED 3 long leg (+)

Step 4: Connect LED 3 short leg (-) back
to battery (-)

Step 5: All 3 LEDs should glow!



Safety First!

Important safety rules:

- Always use LOW voltage batteries only (1.5V or 3V — never mains electricity!)
- Never put wires in a wall socket
- Do not lick or taste batteries
- If a wire feels hot, disconnect immediately
- Always work with a teacher or adult nearby

Electricity is useful but can be dangerous!



Series vs Parallel

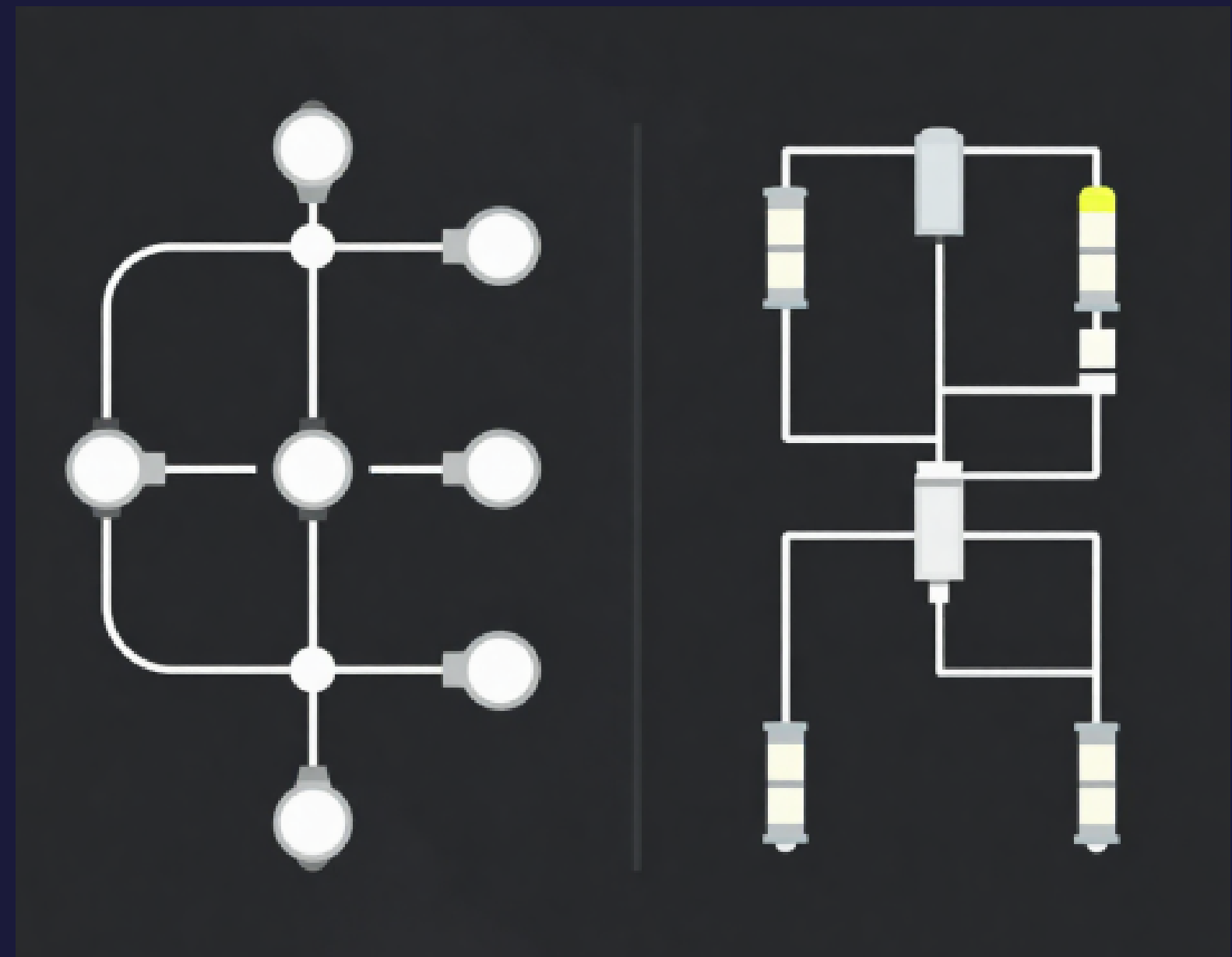
Series Circuit:

- One path for electricity
- LEDs share voltage — dimmer
- One breaks = all stop

Parallel Circuit:

- Many paths for electricity
- Each LED gets full voltage — brighter
- One breaks = others still work

Both are useful in different ways!



Real-Life Examples

Series circuits are all around us!

- Diwali string lights — one bulb out, the whole string goes dark
- Old Christmas tree lights
- Some torch (flashlight) circuits
- Battery cells inside a TV remote

Next time lights go out in a string, you know why — it is a series circuit!



Quick Quiz!

Test what you learned!

1. In a series circuit, how many paths does electricity have?
2. What happens if one LED is removed from a series chain?
3. Do LEDs get brighter or dimmer when we add more in series?
4. Name one real-life series circuit.



Fun Fact!

Did you know?

The first LED was invented in 1962!
It could only glow red.

Today we have LEDs in every colour —
red, green, blue, white, and even
ultraviolet!

LEDs use much less electricity than
old bulbs. That is why they last longer
and keep our planet greener!



What Did We Learn?

Today we discovered:

- A circuit is a complete loop for electricity
- LEDs are tiny lights that use less power
- A series circuit has only ONE path
- LEDs in series share voltage – dimmer
- If one LED breaks, all go out!
- Parallel circuits keep others working

You are now a circuit explorer!

